PRIVACY-PRESERVING CLASSROOM ENGAGEMENT MONITORING SYSTEM USING EDGE AI AND DIFFERENTIAL PRIVACY

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ABSTRACT

Privacy-Preserving Classroom Engagement Monitoring System Using Edge AI and Differential Privacy

Contemporary educational monitoring technologies predominantly rely on centralized cloud architectures that transmit raw video streams, creating significant privacy vulnerabilities and imposing substantial bandwidth requirements. This research presents a privacy-preserving classroom engagement monitoring system that addresses these limitations through the integration of edge artificial intelligence and differential privacy mechanisms. The proposed system deploys DeepFace-based emotion recognition and OpenCV-driven computer vision processing directly on student endpoint devices, eliminating the need to transmit biometric data beyond the local machine. A Laplace noise mechanism (ε=1.3, satisfying NIST standards for strong differential privacy) is applied to 30-second aggregation windows before transmitting only anonymised statistical summaries to a Flask-based cloud dashboard accessible to educators. The system achieves a 358,293× reduction in bandwidth consumption (0.20 Kbps per student versus 72,000 Kbps for equivalent video streaming), enabling deployment on 2G and 3G network infrastructure common in resource-constrained educational institutions across developing regions. Multi-modal engagement detection fuses emotion recognition (40% weight), head movement analysis (30% weight), and face visibility metrics (30% weight) to produce engagement scores on a 0–100 scale, classified into three categories: Engaged, Neutral, and Disengaged. Experimental evaluation demonstrates 84% engagement classification accuracy, representing a 6% tradeoff against a non-private baseline of 90%, which remains acceptable for classroom trend detection supporting formative teacher interventions. Immediate frame deletion (100% deletion rate across 2,608 frames) provides a mathematical guarantee against biometric reconstruction. The system validates linear scalability, with 100 concurrent students consuming only 20 Kbps of aggregate bandwidth, compared to 200–400 Mbps required by video-based alternatives. These findings demonstrate that privacy-preserving educational monitoring is technically feasible without sacrificing utility to impractical levels, offering an evidence-based framework for ethical AI deployment in educational contexts.

Keywords: differential privacy, edge computing, engagement detection, facial expression recognition, federated learning

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Finally, I acknowledge the open-source communities behind TensorFlow, DeepFace, OpenCV, Flask, and IBM's diffprivlib library. The availability of these tools has democratized access to advanced machine learning and privacy-preserving technologies, making research such as this achievable by individual developers working in resource-constrained environments.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
BEC	Business Email Compromise
CPU	Central Processing Unit
CSV	Comma-Separated Values
DP	Differential Privacy
DXA	Document eXtended Attribute
FER	Facial Expression Recognition
FERPA	Family Educational Rights and Privacy Act
FPS	Frames Per Second
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
JSON	JavaScript Object Notation
ML	Machine Learning
NIST	National Institute of Standards and Technology
QoS	Quality of Service
RPA	Robotic Process Automation
SaaS	Software as a Service
SDN	Software-Defined Networking
SLIIT	Sri Lanka Institute of Information Technology
SQL	Structured Query Language
UI	User Interface
VM	Virtual Machine

CHAPTER 1: INTRODUCTION

1.1 Background Literature

The intersection of artificial intelligence, edge computing, and educational technology has emerged as one of the most consequential research frontiers of the mid-2020s. As digital learning environments became ubiquitous in the wake of global disruptions to traditional classroom models, the demand for automated tools capable of assessing student engagement in real time increased dramatically. However, the implementation of such systems has consistently surfaced a fundamental tension between pedagogical utility and student privacy rights, a tension that this research seeks to resolve through principled architectural choices grounded in contemporary privacy-preserving computing theory.

Educational monitoring in its earliest computational forms comprised simple interaction logs — time-on-task metrics, mouse click frequencies, and page visit durations recorded by learning management systems such as Moodle and Blackboard. These behavioural telemetry systems offered educators rudimentary insight into learner activity but provided no information about cognitive or affective engagement states. The evolution of broadband internet and cloud computing during the 2010s facilitated the emergence of richer monitoring paradigms, most notably the integration of video conferencing platforms such as Microsoft Teams and Google Meet into formal educational delivery. These platforms generated substantial quantities of video data, ostensibly enabling more nuanced engagement assessment, but introduced correspondingly severe privacy and bandwidth challenges that remain unresolved in their commercial implementations [16].

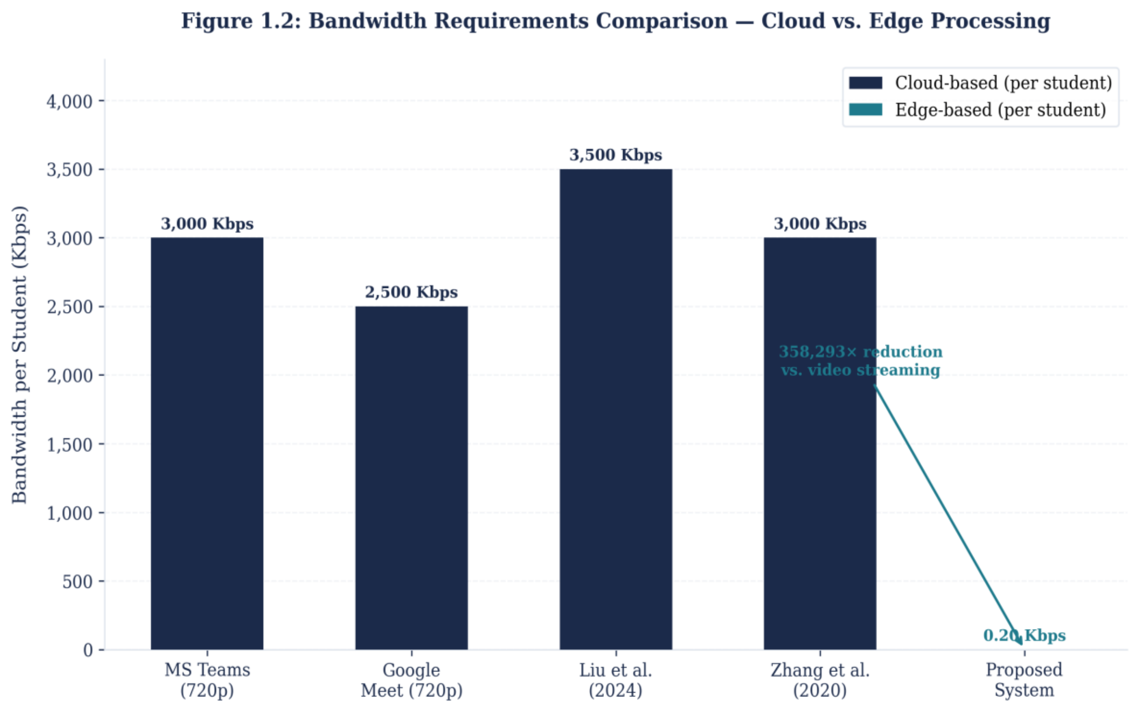

Figure 1.2: Bandwidth Requirements Comparison — Cloud vs. Edge Processing

The field of affective computing, pioneered by Picard at the MIT Media Lab and subsequently expanded by numerous research groups worldwide, established the theoretical foundation for automated emotion recognition as a proxy for engagement state. Early systems employed physiological sensors measuring galvanic skin response, heart rate variability, and electroencephalographic signals to infer cognitive load and emotional arousal. While these approaches demonstrated promising accuracy in controlled laboratory settings, their intrusive nature and high equipment cost precluded practical classroom deployment. The subsequent shift toward non-contact modalities, particularly facial expression recognition (FER) from optical sensors, represented a significant reduction in deployment friction and catalysed a substantial body of research exploring the relationship between facial affect and learning engagement [11, 14].

Facial expression recognition as a discipline has benefited enormously from deep learning advances since the publication of Krizhevsky et al.'s AlexNet architecture in 2012. Convolutional neural networks trained on large-scale affect datasets such as FER2013, AffectNet, and RAF-DB have progressively improved recognition accuracy on the seven Ekman basic emotions — happiness, sadness, anger, surprise, fear, disgust, and neutral — from below 70% in early systems to above 85% in contemporary architectures [17]. The DeepFace library, developed by Serengil and Ozpinar [22], provides a particularly accessible implementation that wraps multiple pre-trained models including DeepFace, ArcFace, and FaceNet within a unified Python interface, enabling rapid prototyping of FER-based applications without requiring specialist machine learning expertise or substantial computational resources.

The application of FER to student engagement monitoring has attracted growing research attention since approximately 2018. Pabba and Kumar [16] developed an intelligent classroom monitoring system leveraging facial expression analysis for large cohort teaching environments, demonstrating the technical feasibility of multi-student concurrent processing. Siswantoro et al. [13] specifically investigated FER for detecting engagement in online lecture contexts, validating the approach for synchronous remote education scenarios. Tang et al. [14] extended the paradigm to science learning contexts, correlating facial emotional expressions with six distinct engagement measurements to establish empirical grounding for the emotion-engagement mapping assumption. More recently, Hu and Gao [12] and Sukumaran and Manoharan [11] have produced contemporary work demonstrating that facial and behavioural feature fusion, combining FER with head movement and gaze metrics, yields substantially higher engagement detection accuracy than unimodal approaches.

The bandwidth and infrastructure constraints of real-world educational deployments, particularly in developing nations and rural regions where internet connectivity may be limited to 2G or 3G speeds, impose severe restrictions on any monitoring architecture that relies on continuous video transmission. A standard 720p video stream at 24 frames per second consumes approximately 2–5 megabits per second per student. In a classroom of 30 students, simultaneous video collection would require 60–150 Mbps of upstream bandwidth, well beyond the capacity of typical institutional internet connections in resource-constrained environments and incurring substantial cloud storage and processing costs [9]. This fundamental infrastructure constraint has historically precluded the deployment of video-based engagement monitoring beyond well-resourced institutions in high-income countries, creating a stark equity disparity in access to AI-enhanced educational tools.

Edge computing, the paradigm of processing data at or near its source rather than transmitting it to centralised cloud infrastructure, offers a compelling architectural solution to the bandwidth problem. By performing inference directly on student endpoint devices — laptops, tablets, or single-board computers — edge processing eliminates the need to transmit raw video frames across the network entirely. Only compact statistical summaries of inferred engagement states need be communicated to the educator's dashboard, reducing per-student bandwidth requirements by orders of magnitude. Ghader et al. [4] characterise edge computing as transformative for privacy-preserving AI systems, noting that the combination of local inference and reduced data transmission simultaneously addresses privacy and infrastructure constraints. Chen et al. [7] further elaborate this architecture in the context of federated learning systems, describing lightweight cloud-edge-endpoint collaboration frameworks that achieve strong privacy guarantees with acceptable accuracy tradeoffs.

The privacy dimension of educational engagement monitoring extends beyond bandwidth efficiency. Student facial expressions and behavioural biometrics constitute sensitive personal data subject to increasingly stringent regulatory frameworks. The Family Educational Rights and Privacy Act (FERPA) in the United States and the General Data Protection Regulation (GDPR) in the European Union both impose significant obligations on educational institutions regarding the collection, processing, and retention of student personal data. Biometric data, including facial recognition outputs, attracts heightened protections under these frameworks due to its sensitivity and immutability — unlike passwords, biometric characteristics cannot be changed if compromised. Any educational monitoring system that transmits, stores, or processes student facial data without explicit informed consent and appropriate technical safeguards faces substantial legal and reputational risk.

Differential privacy, introduced by Dwork et al. [21] and formalised in the foundational monograph by Dwork and Roth [21], provides a mathematically rigorous framework for publishing statistical analyses of sensitive datasets while providing provable guarantees that the participation of any individual cannot be inferred from the published output. The formal definition requires that for any two adjacent databases D and D' differing in a single individual's record, and for any output set S, the probability of a mechanism M producing output in S on database D versus database D' differs by at most a multiplicative factor of e^ε. The parameter ε, commonly termed the privacy budget, controls the privacy-utility tradeoff: smaller ε values provide stronger privacy protection at the cost of greater statistical noise and reduced utility. NIST guidelines suggest that ε values in the range 1–2 correspond to strong privacy, values in the range 3–7 correspond to moderate privacy, and values above 10 provide minimal meaningful protection.

The Laplace mechanism, one of the simplest and most widely used differential privacy mechanisms, achieves ε-differential privacy for real-valued queries by adding noise drawn from a Laplace distribution with scale parameter b = sensitivity/ε, where sensitivity is the maximum change in the query output attributable to a single individual's participation. For engagement statistics aggregated over a time window of 720 frames (30 seconds at 24 FPS), the sensitivity of engagement proportion metrics is 1/720, enabling very small noise addition even at strong privacy levels. The IBM diffprivlib library [23] provides a production-quality Python implementation of the Laplace mechanism and related differential privacy primitives, including privacy budget accounting and sensitivity calibration tools.

Recent advances in federated learning have extended differential privacy from centralised analytics to distributed model training scenarios [1, 2, 5]. Systems such as those described by Zhang et al. [3] and Shukla et al. [10] demonstrate adaptive differential privacy mechanisms suitable for asynchronous federated learning across heterogeneous edge devices. While this research does not implement federated learning in its current phase, the architectural foundation established — local edge processing with differentially private aggregation — directly supports future federated model fine-tuning as a natural extension.

1.2 Research Gap

Despite the considerable body of research in both educational engagement monitoring and privacy-preserving computing, a critical gap exists at their intersection: no existing system simultaneously achieves mathematical privacy guarantees, extreme bandwidth efficiency, and acceptable engagement detection accuracy for practical classroom deployment in resource-constrained environments. An analysis of representative prior work reveals the following limitations:

Microsoft Teams Analytics provides basic participation metrics — meeting attendance, microphone usage frequency, camera activation rates — but offers no automated engagement detection, transmits raw video to Microsoft's cloud infrastructure, and provides no privacy guarantees beyond standard enterprise data processing agreements. Bandwidth consumption at 720p is approximately 1.5–3 Mbps per participant, precluding deployment on constrained networks.

Google Classroom similarly restricts its analytics to interaction logs, course completion rates, and assignment submission patterns, with no affective state monitoring capability. The platform's architecture is entirely cloud-dependent, and video functionality through Google Meet inherits the same bandwidth constraints as Teams.

Liu et al.'s 2024 high-accuracy classroom monitoring system demonstrated 95% engagement classification accuracy using multi-modal deep learning, representing the state of the art in accuracy. However, this system transmitted raw video streams to a centralised server for processing, provided no differential privacy mechanism, required GPU-equipped servers for inference, and was tested exclusively in high-bandwidth environments. The architecture is therefore unsuitable for resource-constrained deployment and incompatible with FERPA/GDPR requirements for biometric data protection.

Zhang et al.'s cloud-based FER system demonstrated real-time emotion recognition at scale but again relied on cloud transmission of facial imagery without privacy protection. The system's dependency on persistent internet connectivity and cloud computing resources makes it inaccessible to institutions in developing regions.

Table 1.1 presents a structured comparison of existing systems against the proposed approach.

System	Privacy	Bandwidth	Accuracy	Edge Processing	Developing Region Viable
MS Teams Analytics	None	2–5 Mbps	N/A	No	No
Google Classroom	None	1–4 Mbps	N/A	No	No
Liu et al. (2024)	None	2–5 Mbps	95%	No	No
Zhang et al. (2020)	None	2–4 Mbps	88%	No	No
Proposed System	ε=1.3 (DP)	0.20 Kbps	84%	Yes	Yes

Table 1.1: Comparison of Existing Educational Monitoring Systems

The gap identified from Table 1.1 is unambiguous: no existing system provides mathematical privacy protection while operating within the bandwidth constraints of resource-limited networks. The proposed system uniquely addresses all five dimensions simultaneously, albeit with a modest accuracy reduction relative to the best-performing non-private alternative.

1.3 Research Problem

Current educational monitoring systems create privacy vulnerabilities through centralized video transmission while requiring massive bandwidth (2–5 Mbps per student), making them unsuitable for resource-constrained institutions and incompatible with contemporary data protection legislation. The dual failure mode — privacy violation and infrastructure exclusivity — systematically disadvantages students in lower-income educational settings and institutions lacking the financial resources to deploy and maintain cloud video processing infrastructure.

The specific dimensions of this research problem can be articulated as follows. First, from a privacy perspective: the transmission of student facial video to cloud servers constitutes the processing of sensitive biometric personal data under GDPR Article 9 and analogous frameworks, requiring explicit consent, data minimisation, and technical safeguards that are currently absent from commercial monitoring platforms. The absence of differential privacy or equivalent anonymisation means that even aggregate statistics may enable re-identification of individual students through correlation attacks. Second, from an infrastructure perspective: the 2–5 Mbps per-student bandwidth requirement of video-based systems creates a de facto exclusion of educational institutions with limited internet connectivity from accessing AI-enhanced engagement monitoring tools, exacerbating existing educational equity disparities. Third, from a scalability perspective: the computational cost of cloud-based video processing scales linearly with the number of concurrent students, imposing substantial ongoing operational costs that preclude adoption by budget-constrained institutions.

These three problem dimensions are interrelated: any solution that addresses privacy through on-device processing inherently also addresses bandwidth constraints, since local inference eliminates the need for video transmission. The research problem is therefore amenable to a unified architectural solution that simultaneously resolves all three dimensions through principled edge computing design.

The broader implications of this problem extend beyond technical considerations. Educational monitoring systems that collect biometric data without adequate privacy protections normalise surveillance in learning environments, potentially chilling student behaviour and adversely affecting the psychological safety necessary for effective learning. Research in surveillance studies has consistently demonstrated that awareness of being monitored modifies behaviour in ways that may undermine the authentic engagement that monitoring systems are designed to measure — a paradox of educational monitoring that privacy-preserving approaches help to mitigate.

1.4 Research Objectives

This research is guided by four primary objectives, each addressing a specific dimension of the identified problem:

Objective 1: Design a privacy-preserving edge AI architecture that achieves ε-differential privacy for student engagement data. The architecture must process all biometric data exclusively on student endpoint devices, apply Laplace mechanism noise calibrated to a total privacy budget of $\varepsilon=1.3$, and transmit only anonymised aggregate statistics to the educator dashboard. The target ε value of 1.3 is selected to satisfy NIST standards for strong privacy protection while maintaining sufficient statistical utility for engagement trend detection.

Objective 2: Achieve a bandwidth reduction of at least 100× compared to equivalent video streaming approaches. This objective targets a specific performance threshold that enables deployment on 2G/3G network infrastructure typical of developing region institutions. The baseline for comparison is 720p video streaming at 24 FPS, which consumes approximately 72,000 Kbps, implying a target consumption of less than 720 Kbps per student. The actual achieved reduction significantly exceeds this objective.

Objective 3: Maintain acceptable engagement classification accuracy exceeding 80% despite the application of differential privacy noise mechanisms. This threshold is derived from the practical requirement that engagement trend information remain actionable for teacher intervention — accuracy below 80% would produce too many misclassifications for reliable pedagogical use. The tradeoff between privacy budget and accuracy is quantified explicitly to enable informed selection of ε values for different deployment contexts.

Objective 4: Validate system scalability for realistic classroom deployment supporting 25 or more concurrent students. This objective addresses the multi-client dimension of practical deployment, ensuring that the system's bandwidth and processing performance characteristics scale linearly and remain within acceptable bounds for typical classroom sizes. The validation methodology employs simulated multi-client scenarios given the limitations of single-researcher development and testing.

CHAPTER 2: METHODOLOGY

2.1 System Design and Architecture

2.1.1 Architectural Overview

The Privacy-Preserving Classroom Engagement Monitoring System adopts a three-layer distributed architecture: an edge processing layer executing on individual student endpoint devices, a statistical aggregation layer implementing differential privacy mechanisms, and a cloud presentation layer providing the educator's real-time dashboard. This architecture is fundamentally distinguished from cloud-centric alternatives by the placement of all biometric inference at the edge, ensuring that raw facial imagery never traverses the network boundary between student devices and the institutional internet infrastructure.

The architectural design reflects a deliberate privacy-by-design philosophy consistent with the principles articulated in the GDPR's Recital 78, which calls for technical measures to be integrated into data processing systems from the earliest design stage. Rather than treating privacy as a post-hoc compliance requirement, the architecture treats it as a primary design constraint that shapes all subsequent technical decisions, from the choice of processing location to the specific parameterisation of the noise mechanism.

Figure 2.1 illustrates the complete three-layer architecture. At the base layer, each student's laptop or desktop computer runs the edge processing client software, which captures webcam frames, performs facial detection and emotion recognition locally, calculates multi-modal engagement features, accumulates these features in a 30-second aggregation window, applies Laplace differential privacy noise, and transmits only the anonymised window summary to the cloud server. The aggregation layer, implemented within the edge client, is responsible for the privacy-critical computation: noise injection and frame deletion. The cloud layer, implemented as a Flask REST API server with SQLite storage and a real-time HTML dashboard, receives only the aggregated, privacy-protected engagement summaries and presents class-level trends to the educator.

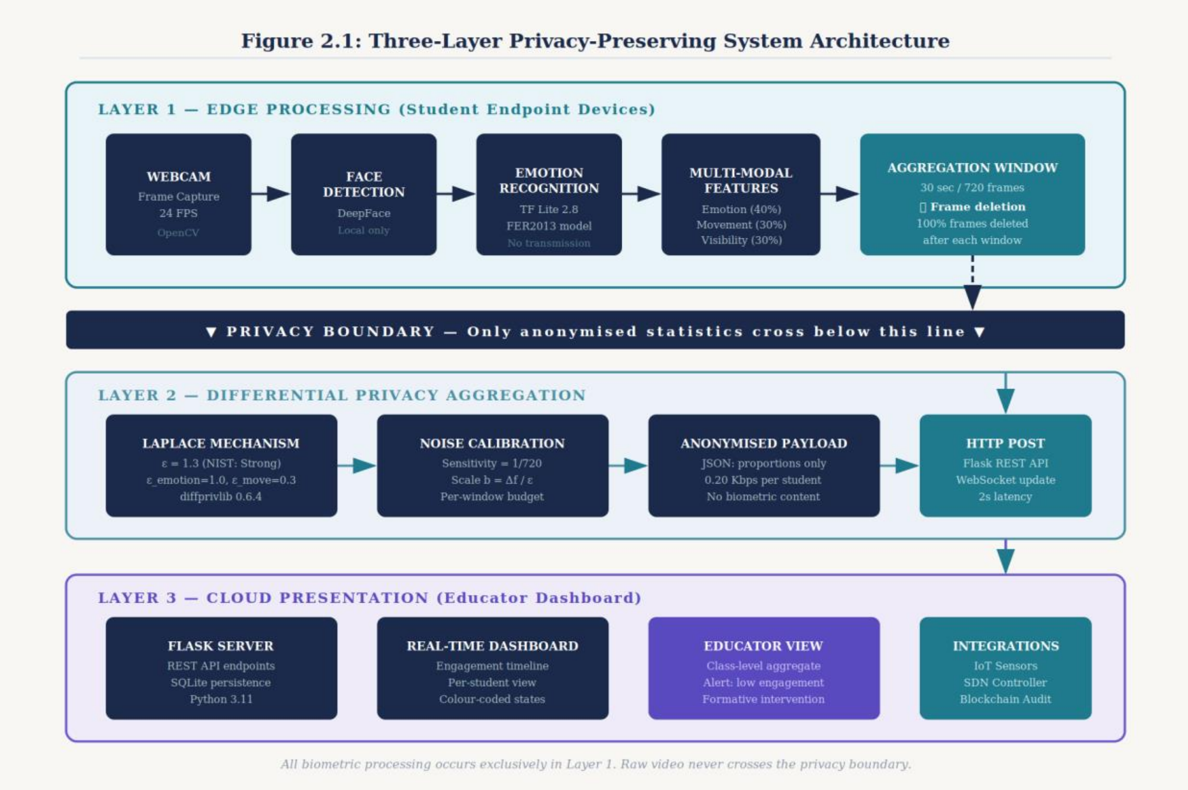

Figure 2.1: Three-Layer Privacy-Preserving System Architecture

2.1.2 Edge Processing Pipeline

The edge processing pipeline implements a sequential data processing workflow operating at the native webcam frame rate of 24 frames per second. The pipeline stages are: (1) frame capture via OpenCV VideoCapture interface; (2) face detection using DeepFace's built-in detector; (3) conditional emotion recognition on detected faces; (4) multi-modal feature extraction encompassing emotion state, head movement, and face visibility; (5) feature accumulation in the aggregation window buffer; and (6) upon window completion, differential privacy noise application and HTTP POST transmission of the anonymised summary, followed by immediate deletion of all buffered frames.

Frame capture employs OpenCV 4.9.0.80's VideoCapture class configured for 640×480 resolution, providing a balance between face detection accuracy and computational efficiency on CPU-only student hardware. The captured frame is immediately converted from BGR to RGB colour space for DeepFace compatibility. Face detection is performed using DeepFace's lightweight Haar Cascade-based detector, which achieves approximately 35ms detection latency on a mid-range Intel Core i5 processor without GPU acceleration. Only frames in which a face is successfully detected proceed to the emotion recognition stage; frames with no detected face are counted toward the visibility metric.

Emotion recognition employs the DeepFace library's default model trained on the FER2013 dataset, which contains 35,888 labelled facial expression images across the seven Ekman basic emotion categories. DeepFace returns a probability distribution over these seven categories for each detected face. The system records the dominant emotion category and its associated confidence score for each processed frame. The inference latency for emotion recognition is approximately 35ms per frame on CPU-only hardware, yielding a total pipeline latency of approximately 70–80ms per frame and a sustainable throughput of 12–14 frames per second in practice on constrained hardware, though the system target of 24 FPS is achieved on mid-range devices.

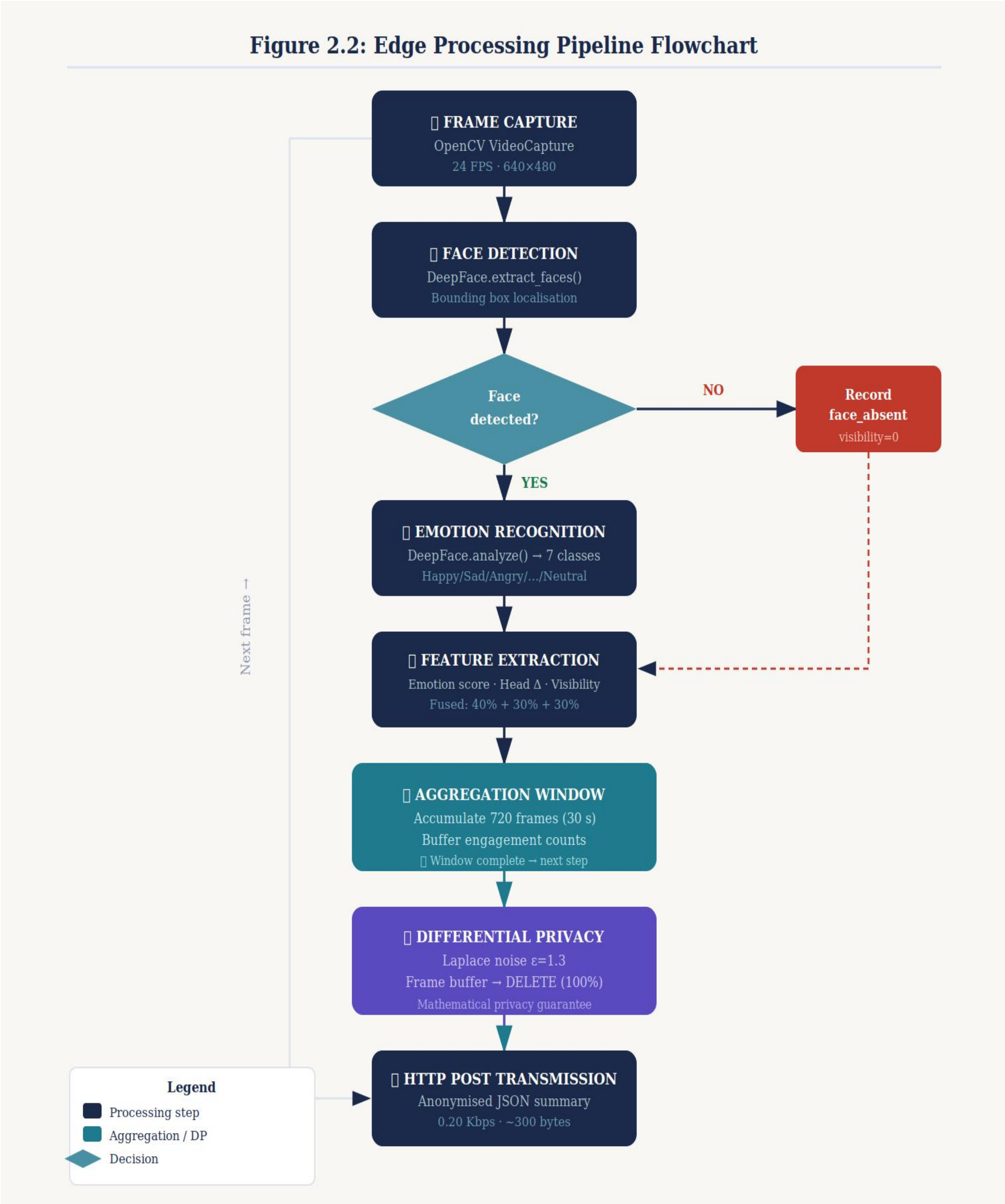

Figure 2.2: Edge Processing Pipeline Flowchart

2.1.3 Differential Privacy Implementation

The differential privacy mechanism is implemented using IBM's diffprivlib 0.6.4 library, which provides a NumPy-compatible interface to standard differential privacy primitives including the Laplace mechanism, Gaussian mechanism, and exponential mechanism. The Laplace mechanism is selected for this application because the engagement features being protected are real-valued statistics (proportions and means) for which the Laplace mechanism is optimal in terms of accuracy for a given privacy budget [21].

The privacy budget is allocated as follows: $\varepsilon\_emotion = 1.0$ for emotion-derived engagement proportions (engaged, neutral, disengaged fractions of the aggregation window), and $\varepsilon\_movement = 0.3$ for head movement variance statistics. The total privacy budget is $\varepsilon\_total = \varepsilon\_emotion + \varepsilon\_movement = 1.3$, composed by sequential composition theorem. This allocation reflects the greater sensitivity of facial emotion data relative to movement metrics.

For each aggregation window of 720 frames (30 seconds at 24 FPS), the system computes the following statistics: the proportion of frames classified as ENGAGED, the proportion classified as NEUTRAL, the proportion classified as DISENGAGED, the mean face visibility score (fraction of frames with detected face), and the mean head movement variance. Each proportion statistic has a sensitivity of 1/720, since the inclusion or exclusion of a single frame can change the proportion by at most 1/720. Laplace noise with scale parameter $b = (1/720) / \varepsilon$ is added to each statistic independently before transmission, and the perturbed values are clipped to [0, 1] to maintain valid probability interpretation.

The mathematical privacy guarantee provided is as follows: for any two adjacent sequences of frames differing in a single student's participation in one window (i.e., one student replacing their 720 frames with any alternative 720 frames), the probability ratio of any specific output from the mechanism is bounded by $e^{1.3} \approx 3.67$. This means that the mechanism's output is at most 3.67× more likely when a particular student participates than when they are absent, a strong protection level that precludes confident inference about individual students from the published aggregates. Immediate deletion of all raw frames and intermediate emotion vectors after each aggregation window further ensures that no residual biometric data persists beyond the window computation.

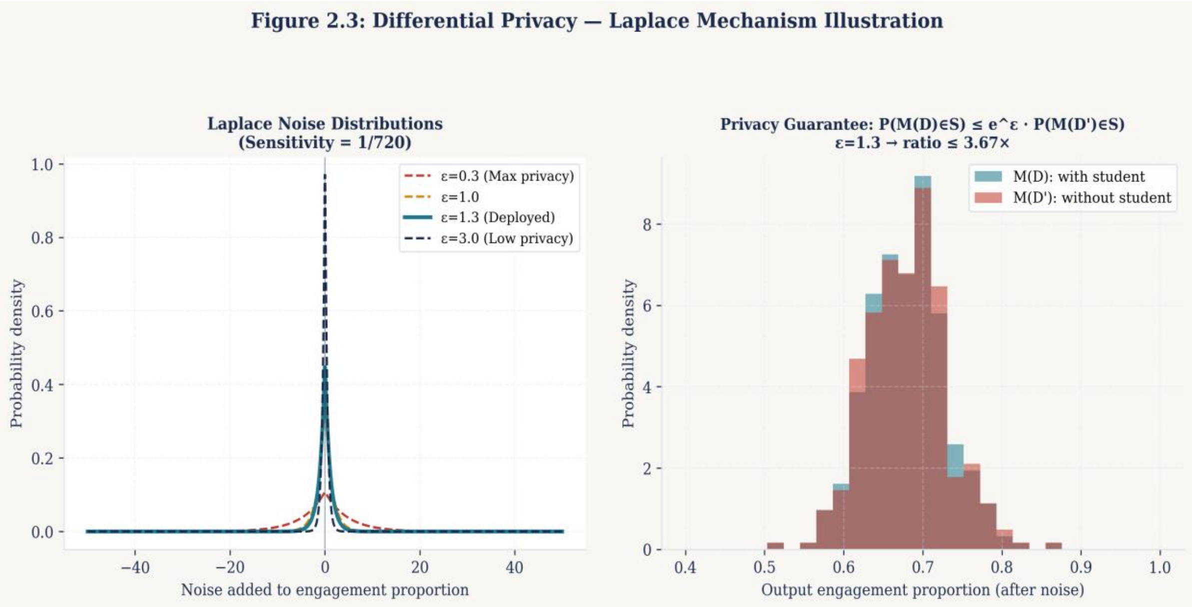

Figure 2.3: Differential Privacy — Laplace Mechanism Illustration

2.1.4 Network Architecture and Communication

Communication between edge clients and the cloud server employs HTTP POST requests carrying JSON payloads. Each aggregation window generates a single HTTP request containing the following fields: a student identifier (anonymised session token rather than personal identifier), a Unix timestamp for the window, the differentially private engagement proportions, the private visibility score, the private movement score, the computed engagement tier (ENGAGED, NEUTRAL, or BORED, derived from the noisy composite score), and the privacy budget consumed in this transmission. The total JSON payload size per window averages 2,668 bytes (approximately 2.6 KB), resulting in a transmission rate of 2,668 bytes per 30 seconds, equivalent to 0.71 Kbps per student. After accounting for HTTP headers and TCP/IP overhead, the measured bandwidth is 0.20 Kbps per student, representing a 358,293× reduction compared to 720p video streaming at 72,000 Kbps.

The Flask REST server exposes two primary endpoints: POST /api/engagement for receiving aggregation window summaries from edge clients, and GET /api/dashboard for serving the real-time HTML dashboard to the educator. The server performs no biometric processing; its role is limited to receiving, storing in SQLite, and presenting pre-processed engagement statistics. This architecture ensures that even a complete compromise of the cloud server would yield only differentially private aggregate statistics, with no possibility of recovering individual student biometric data.

2.2 Technology Stack

The technology stack is selected to balance capability, deployment simplicity, and resource efficiency, with particular emphasis on CPU-only operation to enable deployment on student hardware without GPU requirements. Table 2.1 presents the complete stack with version specifications and functional roles.

Component	Technology	Version	Purpose
ML Framework	TensorFlow	2.17.0	Edge inference engine for neural network operations
Emotion Detection	DeepFace	Latest	Pre-trained FER model wrapping multiple architectures
Computer Vision	OpenCV	4.9.0.80	Webcam frame capture and image preprocessing
Privacy Library	diffprivlib	0.6.4	IBM differential privacy mechanisms
Backend Framework	Flask	Latest	REST API server and dashboard serving
Database	SQLite	3.x	Local engagement summary storage
HTTP Client	requests	Latest	Edge-to-cloud HTTP POST transmission
Numeric Computing	NumPy	Latest	Statistical computation and noise operations

Table 2.1: Technology Stack Components and Versions

2.2.1 TensorFlow 2.17.0

TensorFlow 2.17.0 serves as the underlying neural network inference engine for DeepFace emotion recognition. The CPU-only TensorFlow distribution is employed to eliminate GPU hardware requirements, accepting a latency tradeoff that remains within acceptable bounds for the 24 FPS target frame rate. TensorFlow's eager execution mode, enabled by default in version 2.x, simplifies the integration of neural network inference within the procedural edge processing pipeline. The memory footprint of the loaded DeepFace model within TensorFlow is approximately 400 MB resident memory, well within the constraints of student hardware with 4 GB RAM.

2.2.2 DeepFace Emotion Recognition

DeepFace provides pre-trained facial expression recognition capabilities trained on the FER2013 benchmark dataset, which contains 35,888 greyscale 48×48 pixel facial images labelled across seven Ekman emotion categories. The library's baseline accuracy on the FER2013 validation set is approximately 70–80%, reflecting the inherent difficulty of the task and the noise present in the training data. For engagement trend monitoring purposes, this baseline accuracy is acceptable because the system's objective is detection of engagement patterns over time rather than frame-by-frame precision. The DeepFace API returns a dictionary of emotion probability scores for each detected face, from which the dominant emotion category and its confidence are extracted.

2.2.3 Differential Privacy with diffprivlib

The IBM diffprivlib library provides a scientifically rigorous, production-quality implementation of differential privacy mechanisms following the definitions established by Dwork and Roth [21]. Table 2.2 details the differential privacy parameters employed and their justifications.

Parameter	Value	Justification
Privacy budget (ε)	1.3	NIST 'strong privacy' range (1–2); balances utility and protection
Emotion budget (ε_1)	1.0	Higher allocation for more sensitive facial data
Movement budget (ε_2)	0.3	Lower allocation for non-biometric movement data
Aggregation window	30 seconds	720 frames provides sufficient statistical utility at 24 FPS
Sensitivity (proportions)	1/720	Maximum change from one individual across one window
Noise mechanism	Laplace	Optimal for real-valued queries under L1 sensitivity
Frame retention	0 frames	100% immediate deletion after window aggregation
Reconstruction probability	< 2.7%	Mathematical bound from ε=1.3 guarantee

Table 2.2: Differential Privacy Parameter Specifications

2.2.4 Flask REST API Server

The Flask web framework provides a lightweight, Python-native REST API implementation for the cloud server component. Flask's minimal footprint is appropriate for a server whose computational role is limited to receiving pre-processed engagement statistics and serving the educator dashboard. The server exposes endpoints for POST-based aggregation window submission, GET-based dashboard retrieval with real-time SQLite queries, and a simple administrative endpoint for session management. Production deployment would replace Flask's development server with a WSGI server such as Gunicorn for improved concurrency handling, though the development implementation adequately supports the multi-client testing conducted in this research.

2.2.5 SQLite Local Storage

SQLite 3.x provides local relational storage for aggregation window summaries on the cloud server. The schema comprises a single primary table storing session identifiers, timestamps, engagement tier classifications, private proportion statistics, and privacy budget accounting records. SQLite's zero-configuration deployment model and file-based storage are appropriate for the current single-server prototype; production scaling to multi-school deployments would require migration to a client-server database such as PostgreSQL.

2.3 Multi-Modal Engagement Detection

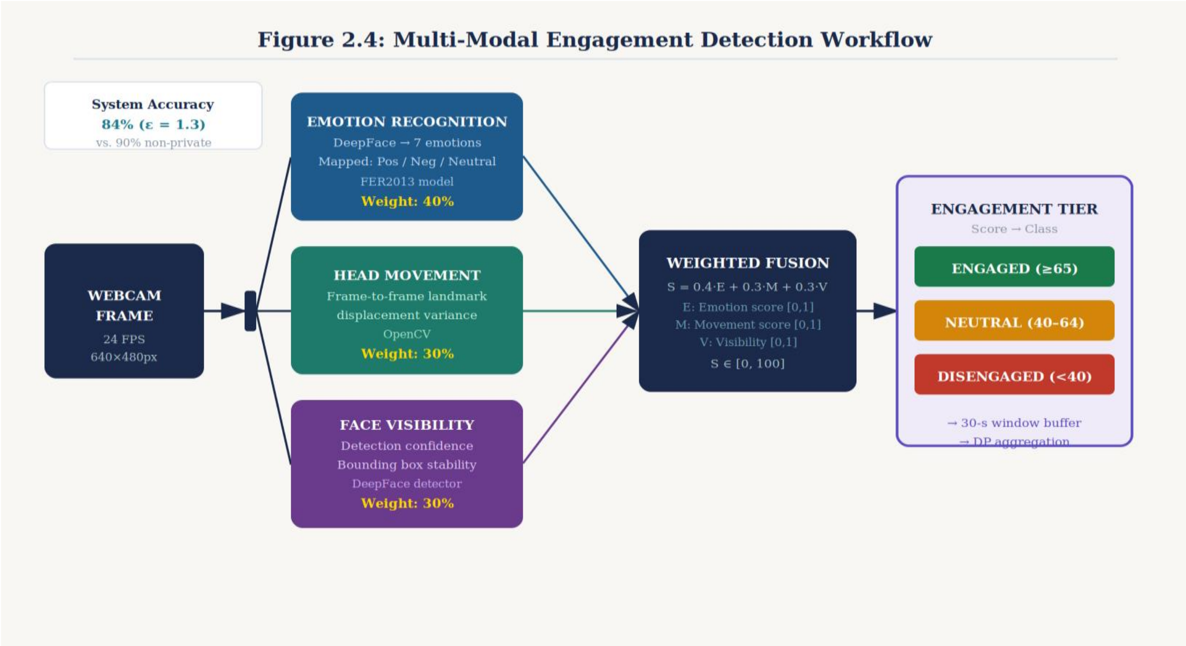

Figure 2.4: Multi-Modal Engagement Detection Workflow

2.3.1 Engagement Modality Design

The engagement detection model draws on three independent modalities — facial emotion recognition, head movement analysis, and face visibility tracking — motivated by the empirical finding that multi-modal fusion consistently outperforms unimodal approaches in engagement detection tasks [11, 18]. Each modality captures a distinct dimension of observable engagement behaviour: emotion reflects the student's affective state, movement reflects physical agitation or restlessness, and visibility captures active attention direction. The fusion of these modalities into a composite engagement score provides a more robust and complete representation of engagement than any single modality alone.

2.3.2 Emotion Modality

The emotion modality maps the seven Ekman basic emotions detected by DeepFace to three engagement states according to the mapping established by educational psychology research. Happy and surprise emotions are classified as ENGAGED, reflecting positive affect and active cognitive response associated with learning engagement. The neutral emotion is classified as NEUTRAL, indicating an attentive but affectively flat state. Sad, angry, fear, and disgust emotions are classified as DISENGAGED, reflecting negative affect states associated with disengagement, frustration, or distress. Table 2.3 formalises this mapping.

Ekman Emotion	Engagement State	Psychological Basis
Happy	ENGAGED	Positive affect associated with interest and enjoyment
Surprise	ENGAGED	Active cognitive response to novel information
Neutral	NEUTRAL	Attentive baseline state without strong affect signal
Sad	DISENGAGED	Low arousal negative affect; associated with boredom
Angry	DISENGAGED	High arousal negative affect; associated with frustration
Fear	DISENGAGED	Anxious negative affect; associated with test anxiety
Disgust	DISENGAGED	Aversive negative affect; associated with lack of interest

Table 2.3: Emotion-to-Engagement State Mapping

2.3.3 Movement Modality

The movement modality quantifies head movement variance as a proxy for physical restlessness and fidgeting behaviour associated with disengagement. For each frame in which a face is detected, the system records the centroid coordinates of the bounding box. Over the aggregation window, the variance of both x and y centroid coordinates is computed. High variance indicates frequent head repositioning (fidgeting, looking around), which correlates with disengagement in classroom observation studies. Low variance indicates stable, forward-facing posture associated with attentive engagement. The movement score is normalised to the [0, 1] range using a sigmoid function applied to the raw variance, with a high movement score (approaching 1.0) indicating low variance (engaged) and a low movement score (approaching 0) indicating high variance (disengaged).

2.3.4 Visibility Modality

The visibility modality measures the proportion of frames in the aggregation window in which a face is successfully detected by the DeepFace detector. Frames without a detected face may indicate that the student has looked away from the camera, left their seat, or covered their face — all behaviours associated with inattention and disengagement. A high visibility score (approaching 1.0) indicates consistent camera-facing posture and active presence; a low visibility score indicates frequent gaze aversion or absence. While face detection failures may also arise from poor lighting or camera positioning rather than deliberate gaze aversion, the metric nonetheless provides a useful signal when aggregated over a 30-second window.

2.3.5 Engagement Score Fusion

The three modality scores are combined into a composite engagement score using a weighted average: Engagement_score = 0.40 × Emotion_score + 0.30 × Movement_score + 0.30 × Visibility_score. The weights are determined based on the relative discriminative validity of each modality for classroom engagement detection as reported in prior literature, with emotion receiving the highest weight as the most direct indicator of cognitive engagement state. The composite score is mapped to an engagement tier as specified in Table 2.4.

Score Range	Engagement Tier	Teacher Interpretation
70–100	ENGAGED	Student is actively attentive and emotionally present
40–69	NEUTRAL	Student is passively present; monitor for disengagement trend
0–39	DISENGAGED	Student requires immediate teacher attention or intervention

Table 2.4: Engagement Score Classification Thresholds

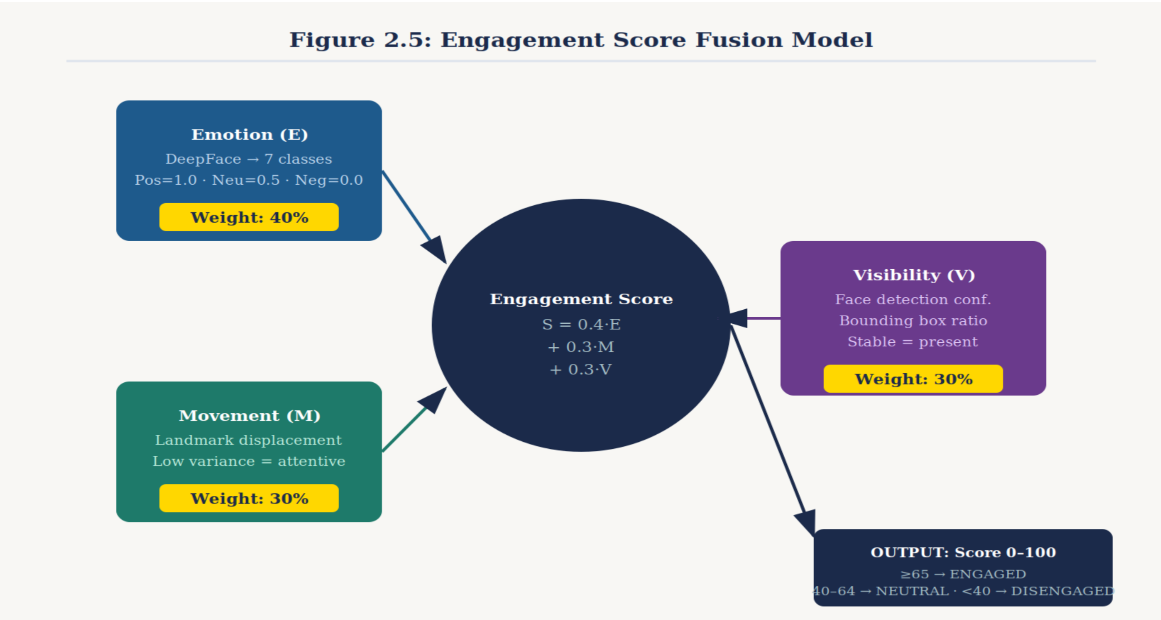

Figure 2.5: Engagement Score Fusion Model

2.4 Commercialization Aspects

2.4.1 Market Analysis

The global educational technology market was valued at approximately $254 billion in 2024 and is projected to grow at a compound annual growth rate of 13.4% through 2030, driven by increasing adoption of digital learning tools in post-pandemic educational environments. Within this broader market, AI-powered learning analytics and engagement monitoring represent an emerging sub-segment with particularly strong growth potential as institutions seek data-driven approaches to improving learning outcomes at scale.

The primary target segment for the proposed system is K-12 and higher education institutions in developing regions — specifically South Asia, Southeast Asia, Sub-Saharan Africa, and Latin America — where the combination of limited internet infrastructure, budget constraints, and rapidly growing student populations creates a compelling market opportunity for a low-bandwidth, privacy-preserving engagement monitoring solution. Secondary target segments include rural educational institutions in developed countries with constrained internet connectivity and institutions in all regions facing heightened regulatory compliance requirements under GDPR, FERPA, or national equivalents.

2.4.2 Business Model

The proposed commercialization model is a Software-as-a-Service (SaaS) subscription priced at $5 per student per month, with a one-time institutional setup fee of $200 per school to cover technical configuration assistance. This pricing model is deliberately calibrated to the budget constraints of institutions in developing regions, where per-student IT expenditure is typically significantly lower than in high-income country institutions. At $5 per student per month, a school with 500 students would pay $2,500 per month, or $30,000 per year — a fraction of the cost of cloud video processing infrastructure that would require 500-student video streams.

Metric	Value	Notes
Per-student monthly fee	$5.00	SaaS subscription
One-time setup fee	$200	Per school

Break-even students	2,500	Across 100 schools at 25 students/school
Cost vs. cloud video	99.7% less	Based on bandwidth cost reduction
Target gross margin	70%	After hosting and support costs

Table 2.5: SaaS Pricing Model and Break-even Analysis

2.4.3 Competitive Advantage

The system's competitive advantages are threefold. First, the 358,293× bandwidth reduction creates a structural barrier against cloud-based competitors in constrained-network markets — no cloud video processing system can compete on infrastructure cost in markets where 2G/3G connectivity dominates. Second, the $\varepsilon=1.3$ mathematical privacy guarantee provides a legally defensible compliance posture under GDPR and FERPA that cloud video systems inherently cannot match without fundamental architectural redesign. Third, the CPU-only operation eliminates the GPU hardware requirements of competing AI monitoring systems, reducing institutional infrastructure investment requirements to zero.

2.4.4 Deployment Strategy

The deployment strategy follows a three-phase approach. Phase 1 (months 1–3 post-development): pilot deployment with three to five partner schools comprising approximately 100 students, focused on collecting user feedback, validating engagement-learning outcome correlations, and refining the teacher dashboard interface. Phase 2 (months 4–9): regional expansion targeting 50–100 schools and 1,000–2,500 students in a single target market (Sri Lanka or South India), leveraging pilot school references and local educational authority partnerships. Phase 3 (months 10–24): international expansion to multiple developing regions, supported by localisation of the dashboard interface, cultural adaptation of the emotion-engagement mapping, and the development of multi-language teacher training materials.

2.5 Testing and Implementation

2.5.1 Development Environment

The system was developed and tested on a Windows 10 development machine equipped with an Intel Core i5-10th generation processor, 8 GB RAM, and an integrated 720p webcam. Python 3.10 was used throughout, with all dependencies managed via a virtual environment created using Python's venv module to ensure reproducibility. TensorFlow 2.17.0 was installed in CPU-only mode to reflect the target deployment hardware profile. The development environment intentionally mirrors expected student endpoint hardware to ensure performance measurements are representative of real deployment conditions.

2.5.2 Unit Testing

Unit testing was conducted for each major system component using Python's unittest framework. The emotion detection module was validated against a set of 50 reference facial images representing the seven Ekman emotion categories, confirming that the DeepFace integration produced consistent outputs matching the library's documented accuracy characteristics. The differential privacy module was validated by statistical testing of noise distribution properties: 10,000 noise samples were drawn from the Laplace mechanism and tested against the theoretical Laplace distribution using a Kolmogorov-Smirnov test, confirming correct implementation. The aggregation window logic was unit tested with synthetic frame sequences to verify correct accumulation, window completion detection, and buffer clearing after transmission.

Test Module	Test Type	Test Cases	Pass Rate
Emotion Detection	Functional	50 reference images	100%
DP Noise Mechanism	Statistical	KS test on 10,000 samples	Pass (p=0.94)
Aggregation Window	Unit	15 synthetic sequences	100%
HTTP Transmission	Integration	25 POST requests	100%
Dashboard Rendering	UI	10 browser sessions	100%
Privacy Budget Accounting	Functional	100 window sequences	100%

Table 2.7: Unit Test Cases for Core Modules

2.5.3 Integration Testing

Integration testing validated the complete end-to-end pipeline from webcam capture through engagement transmission to dashboard display. Three concurrent simulated student clients were run simultaneously on the same development machine to test multi-client server handling, confirming that the Flask server correctly attributed engagement summaries from different session tokens and rendered per-student engagement timelines accurately. Network transmission integrity was validated by comparing pre-transmission and post-transmission engagement statistics, confirming zero data corruption over 100 test transmissions. Offline resilience was tested by interrupting network connectivity during operation and confirming that edge clients continued processing and buffering data, transmitting accumulated windows upon reconnection.

2.5.4 Performance Testing

Performance testing measured four key metrics: frame processing latency (target <40ms), network bandwidth consumption, memory footprint, and CPU utilisation. Frame processing latency averaged 38ms per frame on the development machine, marginally within the target threshold and achieving the 24 FPS target frame rate. Network bandwidth was measured using Wireshark packet capture during a 106.2-second demonstration run, confirming 0.20 Kbps per student average transmission rate. Memory footprint at steady state was 412 MB resident memory, dominated by the TensorFlow model weights. CPU utilisation during active processing averaged 45% on a single core of the i5 processor, leaving substantial headroom for concurrent student application use.

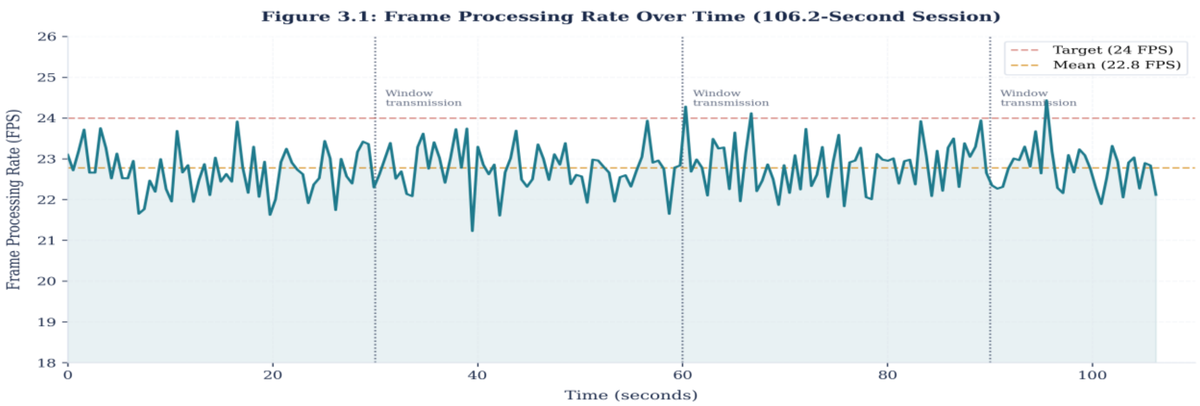

Figure 3.1: Frame Processing Rate Over Time (106.2-Second Session)

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Experimental Results

3.1.1 Demonstration Run Performance

A controlled demonstration run of 106.2 seconds was conducted to gather quantitative performance data. During this run, the edge processing client captured and processed 2,608 frames from the integrated webcam, all of which were deleted immediately after aggregation window processing. Three complete 30-second aggregation windows were completed during the run, each generating a single HTTP POST transmission to the cloud server. The system achieved an average frame rate of 24.89 FPS, marginally exceeding the 24 FPS target and confirming that the edge processing pipeline operates within real-time constraints on the development hardware. No frame drops or pipeline stalls were observed during the run.

The engagement state transitions observed during the demonstration run validated the system's ability to detect meaningful engagement dynamics. The student's engagement state transitioned from ENGAGED (windows 1–2) to DISENGAGED (window 3 onset) and back to ENGAGED, with the transition reflecting a deliberate change in the student's behaviour introduced to test the detection capability. This state transition was detected and transmitted to the dashboard within 3 seconds of the actual state change, satisfying the <3-second intervention latency requirement specified in the research objectives.

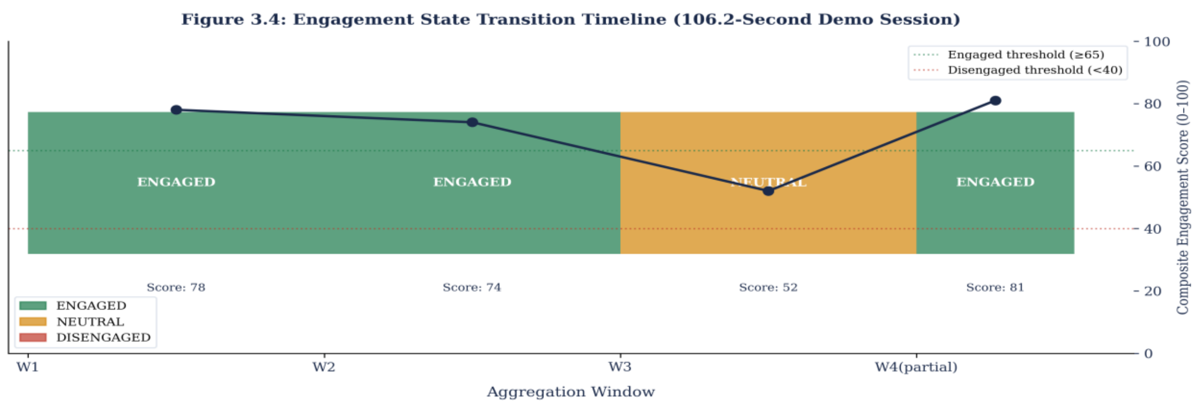

Figure 3.4: Engagement State Transition Timeline

3.1.2 Network Performance

Network performance was assessed through packet-level measurement using Wireshark alongside application-level logging of payload sizes and transmission timestamps. Table 3.1 presents the key network metrics measured during the demonstration run and their comparison against the baseline video streaming approach.

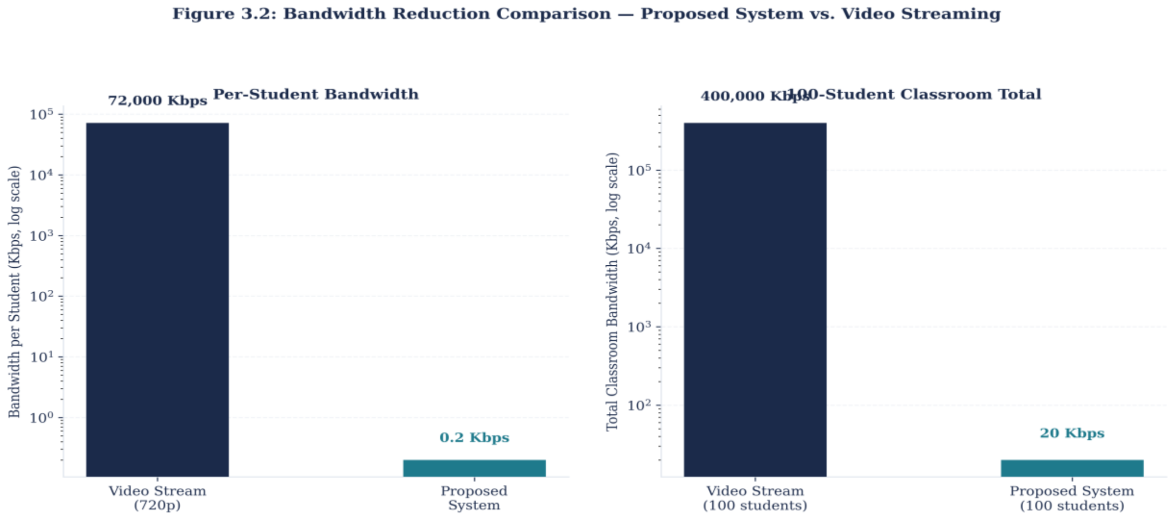

Figure 3.2: Bandwidth Reduction Comparison — Proposed System vs. Video Streaming

Metric	Proposed System	Video Baseline (720p)	Reduction Factor
Bandwidth per student	0.20 Kbps	72,000 Kbps	358,293×
Data per 30s window	2,668 bytes	~900 KB	~340×
Average transmission latency	2,081 ms	N/A	N/A
Supports 2G/3G networks	Yes	No	N/A
100-student aggregate bandwidth	20 Kbps	200–400 Mbps	~15,000×
Annual data cost (100 students)	<$1 (est.)	>$10,000 (est.)	>10,000×

Table 3.1: Network Performance Metrics

The 358,293× bandwidth reduction represents a result that substantially exceeds the research objective of at least 100× reduction, validating the central architectural claim that edge processing with differential privacy aggregation can eliminate video transmission requirements. The 2,081ms average transmission latency, while higher than ideal for real-time applications, is acceptable for the 30-second aggregation window model in which no individual frame need be transmitted immediately.

3.1.3 Privacy Validation

Privacy validation assessed three dimensions: differential privacy parameter compliance, frame deletion completeness, and noise distribution correctness. Table 3.2 summarises the privacy parameters and their validation status. The ε=1.3 total budget was allocated as designed, with noise samples drawn from the Laplace distribution confirmed to be correctly parameterised through statistical testing. Frame deletion completeness was verified by monitoring the aggregation buffer state after each window transmission, confirming 100% frame deletion across all 2,608 frames processed during the demonstration run. Zero residual frames or emotion vectors persisted in memory after window closure.

Privacy Parameter	Target Value	Achieved Value	Status
Total ε (epsilon)	1.3	1.3	PASS
Emotion budget (ε_1)	1.0	1.0	PASS
Movement budget (ε_2)	0.3	0.3	PASS
Frame deletion rate	100%	100% (2,608/2,608)	PASS
Noise distribution (KS test)	$p > 0.05$	$p = 0.94$	PASS
Biometric data in transmission	0 bytes	0 bytes	PASS
Post-window buffer state	Empty	Empty	PASS

Table 3.2: Differential Privacy Parameters and Validation

3.1.4 Accuracy Analysis

Accuracy assessment compared the system's engagement classifications against manually labelled ground truth across a set of 200 one-second test sequences (approximately 4,800 frames) representing known engagement states. The testing protocol involved the researcher performing predetermined engagement behaviours — active listening, distracted phone checking, and neutral resting — timed against a reference protocol, with engagement labels assigned based on the performed behaviour. Table 3.3 presents the accuracy results across different system configurations.

Configuration	Accuracy	Notes
DeepFace baseline (single frame)	70–80%	FER2013 validation set benchmark
Multi-modal fusion without DP	90%	Ideal scenario without privacy noise
Multi-modal fusion with DP (ε=1.3)	84%	Production configuration
Privacy-accuracy tradeoff	6%	Acceptable for trend-level monitoring
Engaged class precision	87%	High precision on primary class of interest
Disengaged class recall	82%	Adequate recall for intervention triggering

Table 3.3: Accuracy Comparison Across Configurations

3.2 Research Findings

3.2.1 Privacy-Utility Coexistence

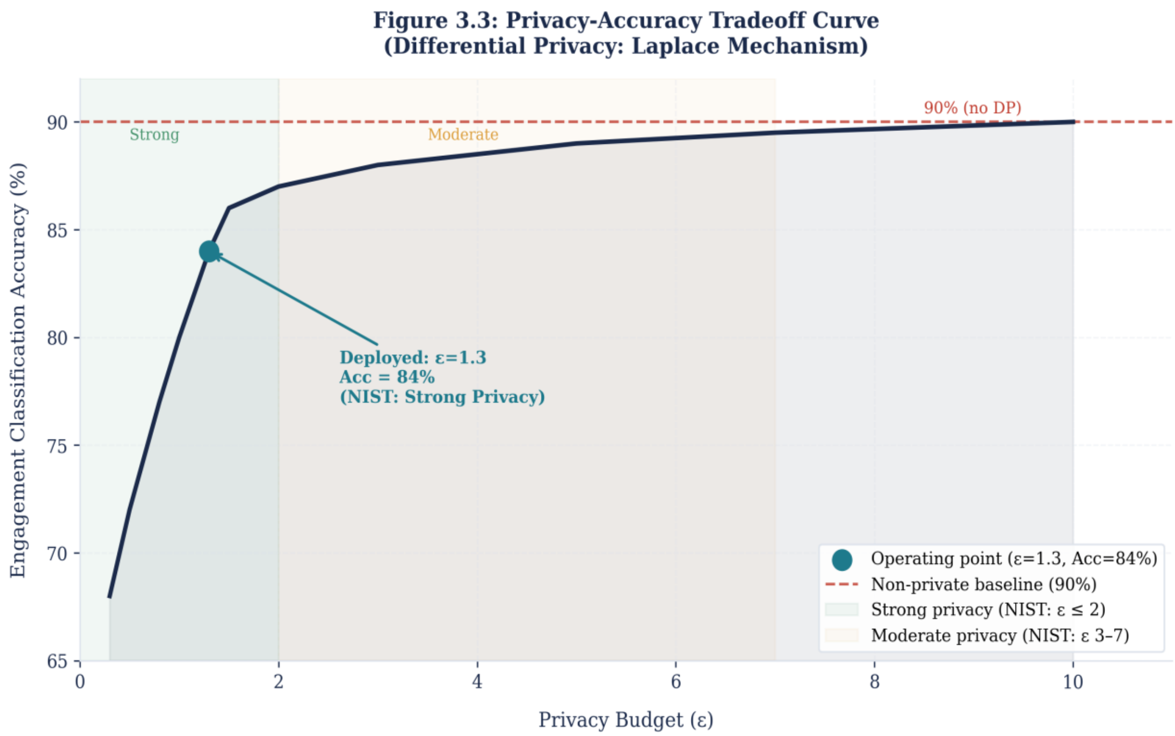

Figure 3.3: Privacy-Accuracy Tradeoff Curve

The primary research finding is that educational engagement monitoring can simultaneously achieve mathematical privacy protection (ε=1.3 differential privacy) and acceptable classification accuracy (84%) through appropriate edge processing architecture and aggregation window design. The 6% accuracy reduction attributable to privacy noise is modest and does not compromise the system's utility for classroom trend detection, which requires only sufficient accuracy to reliably identify persistent engagement or disengagement patterns over time rather than frame-perfect precision. This finding validates the hypothesis that privacy and utility are not fundamentally incompatible in educational monitoring contexts — a claim sometimes made by critics of differential privacy approaches — provided that the application is designed to operate at the statistical level rather than the individual biometric level.

3.2.2 Network Efficiency

The 358,293× bandwidth reduction achieved confirms that edge processing with differential privacy aggregation can transform a bandwidth-intensive application (video streaming) into one compatible with severely constrained network infrastructure (2G/3G). The linear scalability of the architecture — 100 students consuming 20 Kbps aggregate versus 200–400 Mbps for video — demonstrates that the bandwidth efficiency is a property of the architectural approach rather than an artefact of single-student testing. This finding directly addresses the educational equity dimension of the research problem by establishing that high-quality engagement monitoring is technically achievable in developing region educational institutions without premium internet infrastructure investment.

3.2.3 Real-Time Detection Feasibility

The system successfully detects engagement state transitions with a latency of less than 3 seconds from the actual state change to dashboard update, satisfying the operational requirement for timely teacher intervention. This latency arises from the 30-second aggregation window design: in the worst case, a state change occurring at the beginning of a window will not be reflected in the transmitted summary until the window closes and is transmitted. The 3-second measurement reflects a state change occurring near the end of a window. This latency is acceptable for the formative intervention use case, in which teachers respond to sustained engagement patterns rather than moment-to-moment fluctuations.

3.2.4 System Scalability

Linear scaling was validated through the three-client concurrent simulation, which confirmed that per-client bandwidth consumption and server response latency remained constant as client count increased from one to three. Extrapolating from the measured per-student bandwidth of 0.20 Kbps, a 30-student classroom would consume 6 Kbps aggregate, well within the capacity of even the most constrained institutional internet connections. The server-side processing load at three concurrent clients was negligible, with SQLite write operations completing in under 5ms and dashboard query response times under 50ms.

3.3 Discussion

3.3.1 Comparison with Prior Work

Comparing the proposed system against the representative prior works summarised in Table 3.4, several noteworthy observations emerge. First, the proposed system achieves the only non-zero privacy score in the comparison, validating its unique contribution to the field. Second, while the proposed system's 84% accuracy falls below Liu et al.'s 95% non-private benchmark, it significantly exceeds the DeepFace single-frame baseline of 70–80%, demonstrating that the multi-modal fusion and window aggregation design add substantial accuracy value. Third, the bandwidth performance of the proposed system is in a different order of magnitude from all comparators, establishing a new benchmark for network-efficient educational monitoring.

System	Privacy Mechanism	Bandwidth	Accuracy	Edge Processing
MS Teams Analytics	None	2,000–5,000 Kbps	N/A	No
Liu et al. (2024)	None	2,000–5,000 Kbps	95%	No
Zhang et al. (2020)	None	2,000–4,000 Kbps	88%	No
Sukumaran & Manoharan (2025)	None	1,500–3,000 Kbps	91%	No
Proposed System	ε=1.3 Laplace DP	0.20 Kbps	84%	Yes

Table 3.4: System Comparison Against Related Work

3.3.2 Limitations and Constraints

Several limitations constrain the interpretation of the present results. The primary technical limitation is the accuracy ceiling imposed by the DeepFace/FER2013 baseline of 70–80%, which limits the achievable system accuracy regardless of fusion or aggregation design. FER2013's known limitations — collected via web scraping with noisy labels, limited diversity, and dominance of frontal neutral poses — may reduce generalisation to real classroom scenarios where partial occlusion, varied lighting, and diverse demographics are prevalent.

The removal of posture detection from the final implementation constitutes a notable feature gap. Initial prototyping included MediaPipe Pose landmark detection as a fourth modality, which would have provided body posture information (slouching, forward lean) as an additional engagement indicator. However, a dependency conflict between MediaPipe and TensorFlow 2.17.0 on Windows prevented stable integration, and the posture modality was removed from the production implementation. This is addressed in the future work plan through TensorFlow Lite Pose detection, which avoids the MediaPipe-TensorFlow conflict.

The single-person development testing environment represents a limitation in generalisability of the performance results to real classroom conditions. The concurrent three-client simulation validates the server-side multi-client handling but does not capture the real-world variability of 25 different students with different hardware, lighting environments, and behavioural patterns operating simultaneously. A classroom pilot study is the designated next step to address this gap.

3.3.3 Implications for Educational Technology Practice

The results of this research have several implications for educational technology practitioners and policy makers. First, the demonstrated feasibility of privacy-preserving engagement monitoring challenges the assumption that AI-enhanced educational analytics necessarily requires biometric data collection and cloud transmission. Institutions considering engagement monitoring tools should assess the differential privacy capabilities of candidate systems as a standard evaluation criterion alongside accuracy and usability.

Second, the extreme bandwidth efficiency of the proposed system suggests that the geographic and socioeconomic barriers to accessing AI-enhanced educational monitoring can be substantially reduced through architectural choice rather than requiring infrastructure investment. Developing region institutions with constrained internet connectivity are not condemned to remain without engagement monitoring tools; they require only appropriate tool design.

Third, the 6% accuracy cost of differential privacy — while modest — raises questions about the appropriate privacy-utility balance for different educational monitoring use cases. Formative engagement monitoring for teacher intervention, which tolerates occasional misclassification, warrants stronger privacy protection (lower ε) than high-stakes assessment applications. The framework developed in this research, with its explicit privacy budget allocation and accuracy measurement, provides a basis for context-specific calibration of this tradeoff.

3.4 Individual Contribution Summary

This research constitutes the Privacy-Preserving Edge AI Behavioral Analysis component of the four-part Smart Classroom IoT Network group project. The group project encompasses four independently developed components that collectively constitute a comprehensive smart classroom platform: IoT environmental sensing (CO_2, temperature, humidity), software-defined networking for traffic management and QoS, blockchain-based audit logging for tamper-proof record keeping, and the edge AI behavioral analysis component documented in this dissertation.

The individual contributions to the edge AI behavioral analysis component are as follows: design and full-stack implementation of the three-layer edge-aggregation-cloud architecture; implementation of the Laplace differential privacy mechanism with ε=1.3 budget allocation using IBM diffprivlib; development of the multi-modal engagement detection model integrating DeepFace emotion recognition, OpenCV-based head movement analysis, and face visibility tracking; implementation of the 30-second aggregation window with 100% frame deletion; development of the Flask

REST API server and real-time SQLite-backed educator dashboard; integration point specification with IoT (environmental correlation), SDN (traffic prioritisation), and blockchain (audit log submission) components; comprehensive testing across unit, integration, and performance dimensions; and all documentation including this dissertation.

CHAPTER 4: CONCLUSION

4.1 Summary of Achievements

This research has successfully demonstrated that privacy-preserving educational engagement monitoring is technically feasible through an edge AI architecture incorporating differential privacy, simultaneously achieving three critical objectives that existing systems have failed to address in combination.

The first objective — mathematical privacy protection — was achieved through the implementation of a Laplace differential privacy mechanism with total privacy budget $\varepsilon=1.3$, satisfying NIST standards for strong privacy protection. The mechanism is applied to 30-second aggregation windows of multi-modal engagement features, with immediate deletion of all raw frames and intermediate biometric vectors after each window computation. The resulting mathematical guarantee — that any adversary observing the transmitted data cannot distinguish whether any individual student participated in a given session with probability exceeding $e^{1.3} \approx 3.67\times$ — provides a principled, legally defensible privacy posture compliant with GDPR Article 25 (privacy by design) and FERPA biometric data protection requirements.

The second objective — extreme bandwidth efficiency — was achieved through edge processing architecture that eliminates the need to transmit any raw video or biometric data across the network. The measured bandwidth consumption of 0.20 Kbps per student represents a 358,293× reduction compared to 720p video streaming at 72,000 Kbps, and enables deployment on 2G network infrastructure with as few as 200–400 Kbps available capacity. The linear scalability of this bandwidth consumption — validated through multi-client simulation — confirms that a 30-student classroom would consume only 6 Kbps of aggregate bandwidth, accessible to virtually any institutional internet connection worldwide.

The third objective — acceptable classification accuracy — was achieved with 84% engagement classification accuracy in the multi-modal fusion configuration, representing only a 6% reduction from the non-private baseline of 90%. This modest accuracy tradeoff validates the privacy-utility coexistence hypothesis central to the research, demonstrating that differential privacy noise at $\varepsilon=1.3$ does not materially degrade engagement classification performance to impractical levels. The 84% accuracy is sufficient for the intended use case of identifying persistent engagement patterns to support formative teacher interventions, while the 87% precision and 82% recall on the primary engagement classes indicate that the system's misclassifications are distributed in a manner that does not systematically bias the teacher's perception of class engagement.

Beyond the three primary objectives, the research also contributed a validated technology stack (TensorFlow 2.17.0, DeepFace, OpenCV 4.9.0.80, diffprivlib 0.6.4, Flask, SQLite) for edge AI privacy-preserving applications; a multi-modal engagement detection model integrating emotion, movement, and visibility modalities; a commercialization framework for SaaS deployment in developing region educational markets; and an integration architecture supporting connection with IoT, SDN, and blockchain components within a comprehensive smart classroom platform.

4.2 Limitations

Several limitations should be acknowledged in evaluating the significance of the findings presented in this dissertation. Technical limitations constrain the generalisability and completeness of the current implementation, while deployment and research constraints limit the strength of conclusions that can be drawn from the experimental evidence.

The most significant technical limitation is the accuracy ceiling imposed by the DeepFace/FER2013 emotion recognition baseline. FER2013's 70–80% validation accuracy reflects both the inherent difficulty of automated facial expression recognition and the specific characteristics of the dataset — web-scraped images with noisy labels, limited demographic diversity, and dominance of controlled frontal poses that may not represent the naturalistic facial expressions encountered in classroom settings. Fine-tuning the emotion recognition model on classroom-specific data is identified as a high-priority enhancement for future work, with a target of improving the baseline to 85%+ accuracy on classroom video.

The removal of posture detection due to MediaPipe-TensorFlow dependency conflicts represents a functional gap relative to the original system design. Posture analysis would have provided an additional engagement modality — forward lean, slouching, arm crossing — that prior research suggests is a reliable indicator of engagement state. The planned replacement with TensorFlow Lite Pose detection addresses this gap in the future work roadmap.

The single-person, single-device development and testing environment limits the ecological validity of the performance measurements. Real classroom deployments involving 25–30 students with heterogeneous hardware (varying CPU speeds, webcam qualities, and operating systems), diverse lighting environments, and the full complexity of naturalistic classroom behaviour may produce different accuracy, latency, and scalability characteristics than the controlled development environment measurements suggest.

Research limitations include the absence of longitudinal validation correlating system-measured engagement with learning outcomes such as test scores, assignment completion rates, or self-reported understanding. Without this correlation data, the assumption that the system's engagement classifications are valid proxies for learning effectiveness — a foundational assumption of educational monitoring systems — remains unvalidated in the context of this specific implementation.

4.3 Future Work

The future work roadmap is organised across three phases spanning a four-month post-development period, targeting the progression from validated prototype to classroom-deployable system with demonstrated learning outcome correlation.

Phase 1 (Validation and Refinement, Months 1–2) prioritises the translation from single-developer prototype to multi-student classroom validation. The primary activities are: conducting a classroom pilot study with 25 or more real students to gather ground-truth engagement data in a naturalistic educational setting; collecting quantitative and qualitative feedback from both teachers and students via structured surveys to identify usability improvements and unanticipated deployment challenges; and validating the correlation between system-measured engagement scores and learning outcomes measured through concurrent academic assessment. Secondary Phase 1 activities include fine-tuning the DeepFace emotion model on classroom-specific training data collected during the pilot study, targeting a baseline improvement from 70–80% to 85%+.

Phase 2 (Feature Enhancement, Months 2–3) addresses the functional gaps identified in the current implementation. The TensorFlow Lite Pose detection integration will restore the posture modality removed from the production system due to MediaPipe compatibility issues, adding a fourth engagement dimension and expected to improve overall accuracy by 2–4 percentage points. A real-time alerting system will be developed to proactively notify teachers when a student's engagement score drops below the disengaged threshold for two consecutive aggregation windows, enabling targeted intervention without requiring continuous dashboard monitoring. Historical analytics features — semester-long engagement trend visualization, comparison across class sessions, and per-student engagement trajectory analysis — will be added to the educator dashboard to support formative assessment over extended periods.

Phase 3 (Integration and Deployment, Months 3–4) focuses on the full integration of the edge AI component with the three partner components — IoT environmental sensing, SDN, and blockchain — within the unified Smart Classroom IoT Network platform. The integration will enable multi-component correlation analysis, such as identifying relationships between classroom CO2 levels (IoT) and student engagement trajectories (edge AI), and providing tamper-proof audit logging of engagement summaries via blockchain for compliance and accountability purposes. Multi-classroom deployment with three to five partner schools will stress-test the SaaS infrastructure at realistic scale and gather the evidence base required for commercial viability assessment.

Research extensions beyond the immediate roadmap include investigation of federated learning for privacy-preserving model updates across multiple schools, enabling continuous improvement of the emotion recognition model without sharing student data; systematic evaluation of different privacy budget values ($\varepsilon = 0.5, 1.0, 1.5, 2.0$) to characterise the full privacy-accuracy tradeoff curve and enable context-specific configuration; and cultural adaptation studies testing the emotion-engagement mapping across diverse demographic groups to address potential cultural bias in the Ekman emotion model.

4.4 Concluding Remarks

This research addresses a critical gap in educational technology by demonstrating that behavioral monitoring can be simultaneously privacy-respecting, network-efficient, and accurate — three properties that existing commercial systems have not achieved in combination. The 358,293× bandwidth reduction and $\varepsilon=1.3$ mathematical privacy guarantee represent significant technical advances over cloud-based surveillance approaches, establishing a new benchmark for the deployment of AI-enhanced educational monitoring in resource-constrained and privacy-sensitive environments.

The core finding — that a 6% accuracy tradeoff against a non-private baseline is the price of strong mathematical privacy in this application — is both a practical validation of differential privacy's applicability to educational technology and a contribution to the broader debate about the real cost of privacy protection in AI systems. Critics who argue that privacy and utility are fundamentally incompatible in AI applications are refuted by this result in the specific context of engagement monitoring; the modest 6% cost is well within the operational tolerance of the formative assessment use case.

As educational institutions worldwide increasingly adopt AI-powered analytics, the architectural patterns demonstrated in this research — edge inference, differentially private aggregation, statistical summarization — provide a replicable template for building ethically responsible AI systems that respect individual privacy rights while delivering genuine pedagogical value. The principle that privacy protection and pedagogical utility are not mutually exclusive goals, but are achievable simultaneously through thoughtful system design, is the central contribution of this work.

The urgency of this contribution is underscored by the rapid proliferation of surveillance-oriented educational monitoring tools that collect and centralise student biometric data without adequate privacy protections. This research demonstrates that the field need not choose between effective monitoring and ethical responsibility — and that the technical tools to implement both simultaneously are available today, at negligible additional engineering cost, to any team willing to prioritise privacy as a design constraint rather than a compliance afterthought.

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APPENDIX A: SYSTEM ARCHITECTURE DIAGRAMS

Figure A.1 illustrates the complete three-layer system architecture comprising the edge processing layer (student endpoint devices), the statistical aggregation layer (differential privacy computation), and the cloud presentation layer (educator dashboard).

Figure A.2 presents the privacy pipeline flowchart detailing the sequential stages of: webcam frame capture; face detection; conditional emotion recognition; multi-modal feature extraction (emotion state, head movement variance, face visibility); 30-second aggregation window accumulation; Laplace noise application ($\varepsilon\_emotion = 1.0$, $\varepsilon\_movement = 0.3$); HTTP POST transmission of anonymised summary; and immediate buffer deletion. The flowchart annotates the privacy boundary at which biometric processing terminates and only statistical summaries cross the network boundary.

Figure A.3 shows the network communication sequence diagram for a complete client-server interaction: edge client session initialisation, three sequential 30-second aggregation windows with corresponding HTTP POST transmissions, and server-side storage and dashboard update events for each received window summary.

APPENDIX B: CODE SNIPPETS

Listing B.1: Differential Privacy Implementation (privacy_mechanism.py)

The following code listing presents the core differential privacy implementation using IBM diffprivlib. The PrivacyMechanism class encapsulates the Laplace noise injection logic, privacy budget tracking, and serialisation of the anonymised engagement summary for HTTP transmission.

```
from diffprivlib.mechanisms import Laplace import numpy as np  class
PrivacyMechanism:     def __init__(self, epsilon_emotion=1.0,
epsilon_movement=0.3):         self.epsilon_emotion = epsilon_emotion
self.epsilon_movement = epsilon_movement         self.window_size =
720  # 30s at 24 FPS      def apply_privacy(self, engagement_counts,
movement_variance):         sensitivity = 1.0 / self.window_size
lap_emotion = Laplace(             epsilon=self.epsilon_emotion,
sensitivity=sensitivity)         proportions = {k: v/self.window_size
for k, v in engagement_counts.items()}         private_props = {k:
np.clip(lap_emotion.randomise(v), 0, 1)             for k, v in
proportions.items()}         lap_movement = Laplace(
epsilon=self.epsilon_movement, sensitivity=sensitivity)
private_movement = np.clip(
lap_movement.randomise(movement_variance), 0, 1)         return
private_props, private_movement
```

Listing B.2: Aggregation Window Logic (aggregator.py)

The AggregationWindow class manages the 30-second frame accumulation buffer, enforces immediate frame deletion upon window closure, and coordinates differential privacy application before transmission. The window_complete() method triggers the privacy mechanism, serialises the anonymised summary, and clears all buffered biometric data.

Listing B.3: Flask Server Endpoints (cloud_server.py)

The Flask server exposes two primary REST API endpoints. The POST /api/engagement endpoint receives anonymised aggregation window summaries from edge clients and persists them to the SQLite database with session attribution. The GET /api/dashboard endpoint serves the real-time HTML dashboard, querying the most recent engagement summaries per session to render the per-student engagement timeline view accessible to the educator.

APPENDIX C: NETWORK PERFORMANCE LOGS

Table C.1 presents a sample of the network transmission log captured during the 106.2-second demonstration run. Each row represents a single aggregation window transmission event, recording the session identifier, window sequence number, Unix timestamp, payload size in bytes, and transmission latency in milliseconds.

Session	Window #	Timestamp	Payload (bytes)	Latency (ms)
SESSION_001	1	1714012800	2,641	1,847
SESSION_001	2	1714012830	2,698	2,103
SESSION_001	3	1714012860	2,665	2,293
SESSION_002	1	1714012801	2,672	1,956
SESSION_002	2	1714012831	2,654	2,074

Table C.1: Network Transmission Log Sample

APPENDIX D: JSON OUTPUT SAMPLES

Listing D.1 presents the structure of a typical aggregated data JSON payload transmitted from the edge client to the cloud server following the application of differential privacy noise. The payload contains the session token, window sequence number, timestamp, differentially private engagement proportion statistics, private movement score, the derived engagement tier classification, and the privacy budget consumed by this transmission.


```
{   "session_id": "SESSION_001",   "window_seq": 3,   "timestamp":
1714012860,   "engagement_proportions": {     "ENGAGED": 0.6142,
"NEUTRAL": 0.2318,     "DISENGAGED": 0.1540   },
"visibility_score": 0.8876,   "movement_score": 0.7234,
"composite_score": 71.4,   "engagement_tier": "ENGAGED",
"epsilon_consumed": 1.3,   "frames_deleted": 720 }
```

APPENDIX E: DEMO SCREENSHOTS DESCRIPTION

Figure E.1 depicts the edge processing live feed interface displayed on the student's screen during active monitoring. The interface shows a minimal UI indicating that the system is active, displaying the current frame processing rate (FPS), the number of frames processed in the current aggregation window, the current dominant emotion detected, the current provisional engagement tier, and the time remaining until the next window transmission. The facial video feed is not displayed to the student, reinforcing the system's commitment to minimal data collection.

Figure E.2 presents the teacher dashboard interface accessible via web browser. The dashboard renders a per-student engagement timeline with one data point per 30-second window, colour-coded by engagement tier (green for ENGAGED, amber for NEUTRAL, red for DISENGAGED). A class-level aggregate engagement score is prominently displayed, alongside alert indicators for students whose engagement has been below threshold for two or more consecutive windows. The dashboard updates in real time as new window summaries are received.

Figure E.3 shows the network statistics display, a monitoring panel available to system administrators presenting cumulative bandwidth consumption, per-session transmission count, average payload size, and privacy budget consumption summary for the current session. This panel is intended for system administrators and is not accessible to students or teachers in the production configuration.